

Limits of AI peer review on real expert review data

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Abstract

We benchmark how frontier LLMs critique, rate, and prioritize research, comparing AI-generated evaluations to human expert assessments from The Unjournal. Human ratings serve as a reference signal—the core goal is measuring and diagnosing AI reasoning quality: understanding calibration, failure modes, and systematic “taste” differences between AI and human evaluators.

Introduction

i Working paper

This is an evolving working paper. Analysis, metrics, and comparisons are under active development.

Peer review is under strain. Reviewers are hard to find, turnaround times are lengthening, and the system costs an estimated \$1.5 billion per year in the United States alone (Aczel, Szaszi, and Holcombe 2021). An influx of AI-generated manuscripts is likely to intensify the bottleneck (Eger et al. 2025). If frontier language models could produce reliable, structured evaluations of research, they might free substantial scientific resources and deliver faster, more detailed feedback.

We test this possibility using The Unjournal’s structured human evaluations as a benchmark. We prompt six frontier LLMs—GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6, and Gemini 2.0 Flash—with the same rubric and guidelines used by human evaluators, then compare the resulting quantitative ratings and qualitative critiques against human evaluations for 60 economics and social-science working papers. We ask whether frontier AI evaluations can reliably approximate human expert judgment, which models perform best across rating criteria and cost tiers, and whether systematic differences reveal characteristic AI preferences over research.

The Unjournal setting is particularly well suited for this comparison. It commissions paid expert evaluations using a structured rubric covering seven percentile criteria with 90% credible intervals plus journal-tier predictions, and publishes the resulting packages openly—reducing classic gatekeeping motives and increasing reviewer effort. The resulting ratings and critiques still exhibit substantial inter-rater variation; accordingly, we treat human evaluations as a high-quality but noisy *reference signal*, not ground truth. The rich, multi-dimensional data allow us to compare the priorities and calibration of humans and AI models across criteria and domains, while the journal-tier predictions provide an external reference point¹ enabling a human-vs-LLM horse

¹These represent verifiable publication outcomes, not statements about the “true quality” of the paper.

race. Finally, The Unjournal’s pipeline of future evaluations allows for clean out-of-training-data predictions, serving as a live testing lab for prospective validation.

A growing literature examines LLM-assisted peer review, error detection, and AI research evaluation; we survey this work in Section . Our approach differs from prior work by focusing on structured, percentile-based quantitative ratings with credible intervals, comparing against published human evaluations rather than using LLM-as-judge, and curating contamination-aware inputs.

Acknowledgements.

This work is a collaboration with [The Unjournal](#), which has published 55+ detailed evaluation packages building open, quantitative evaluation infrastructure for global-priorities-relevant research in economics, policy, and social science. We thank The Unjournal’s evaluation community for generating the structured human assessments that make this benchmarking possible.

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Results

We evaluate 6 frontier LLMs against human expert reviews from [The Unjournal](#) for 45 papers with both human and LLM evaluations. Each model receives the same PDF, system prompt mirroring The Unjournal rubric, and JSON schema requiring a diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. Full methodological details appear in [Data and Methods](#).

Per-paper overview. Figure 1 displays individual human evaluator ratings (green circles with 90% CIs) alongside the GPT-5 Pro LLM rating (orange diamond with CI) for each paper’s overall score. Papers are ordered by human mean rating. Dotted lines mark grand means.

The spread of green circles within each paper column illustrates substantial human inter-rater variability—CIs often span 20–40 points. In most cases the LLM diamond sits comfortably within the range of human opinions, though several papers show the LLM falling outside the human consensus.

Overall ratings. Figure 2 compares overall ratings from GPT-5 Pro and Claude Opus 4.6—the two highest-performing reasoning models—against human mean ratings. Points on the diagonal indicate perfect agreement; distance from the line reflects disagreement.

Both models cluster around the identity line in the 40–80 range but diverge more at the extremes. LLMs tend to compress ratings toward the centre of the scale relative to humans, producing narrower spread—a pattern consistent with alignment training that discourages extreme outputs. Where humans rate a paper very highly or very harshly, the LLM typically pulls toward the middle. Full agreement metrics including Pearson r , Spearman ρ , mean bias, and RMSE for all six models appear in Table 2.

Rank comparison. Figure 3 places human rankings in the centre column with GPT-5 Pro on the left and Claude Opus 4.6 on the right. Each paper appears as a pair of S-curves connecting its human rank to its rank under each LLM. Horizontal curves indicate agreement; curves that climb toward the top indicate the LLM ranked the paper higher (better) than humans did. Colour encodes the rank shift: green when humans rank it higher, orange/purple when the LLM does. The

Figure 1: Per-paper overall ratings (0–100 percentile). Green circles show individual human evaluator midpoints with 90% credible intervals; the orange diamond shows the GPT-5 Pro rating with its CI. Papers are sorted left-to-right by descending human mean. Dotted lines mark grand means for each evaluator type.

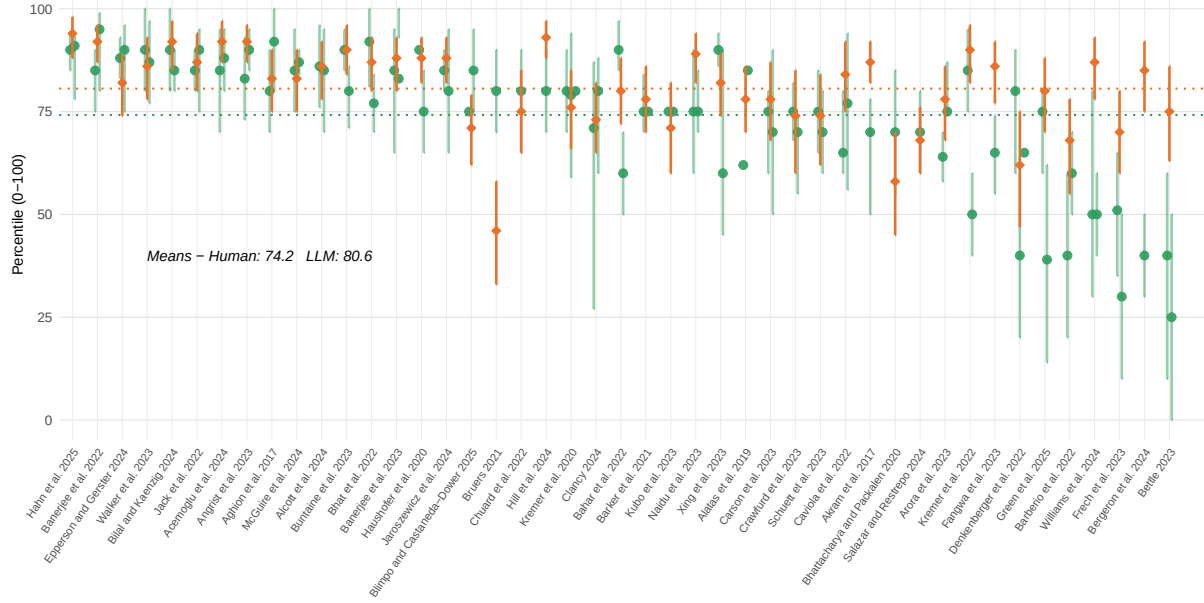
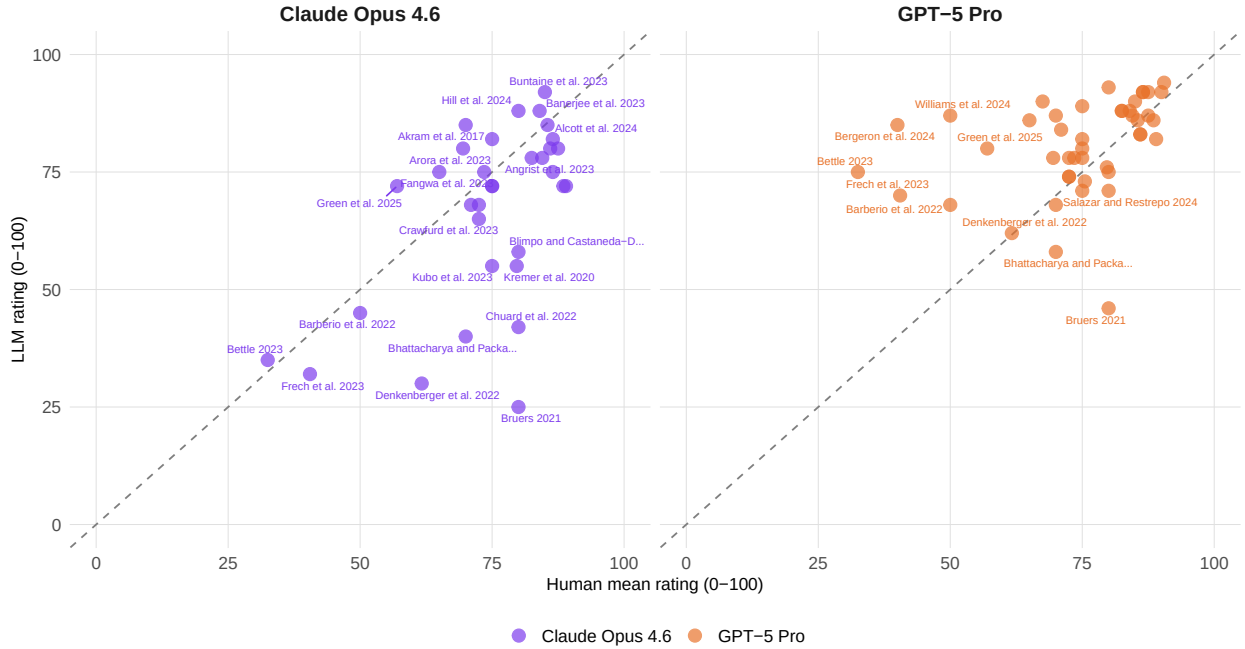
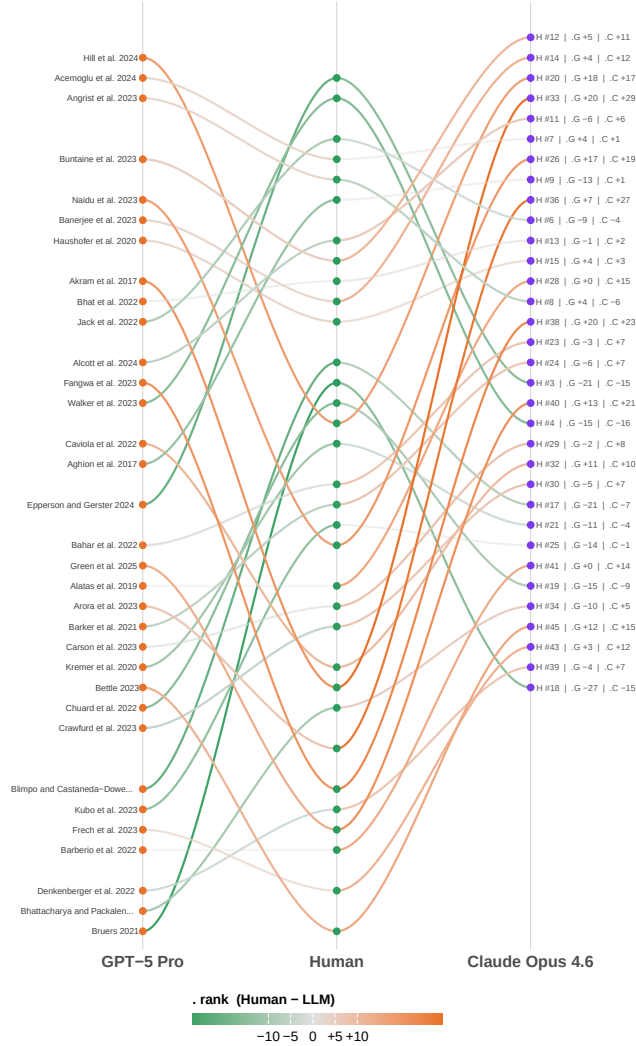


Figure 2: Human mean overall rating (x-axis) vs LLM overall rating (y-axis) for GPT-5 Pro and Claude Opus 4.6. Each point is one paper; the dashed diagonal is the identity line (perfect agreement). Points above the line indicate the LLM rated the paper higher than humans; points below indicate harsher LLM ratings. Paper labels are truncated for readability.



trptych layout makes it easy to spot papers where both LLMs agree with each other but disagree with humans (both curves slope the same way), versus papers where the two LLMs diverge.

Figure 3: Triptych slopegraph of paper rankings (overall criterion). Centre column shows human mean ranking; left and right columns show GPT-5 Pro and Claude Opus 4.6 rankings respectively. Each paper is connected by S-shaped curves; colour intensity encodes the magnitude of rank shift (Δrank). Green = humans rank it higher; warm colour = LLM ranks it higher. Horizontal curves indicate agreement.

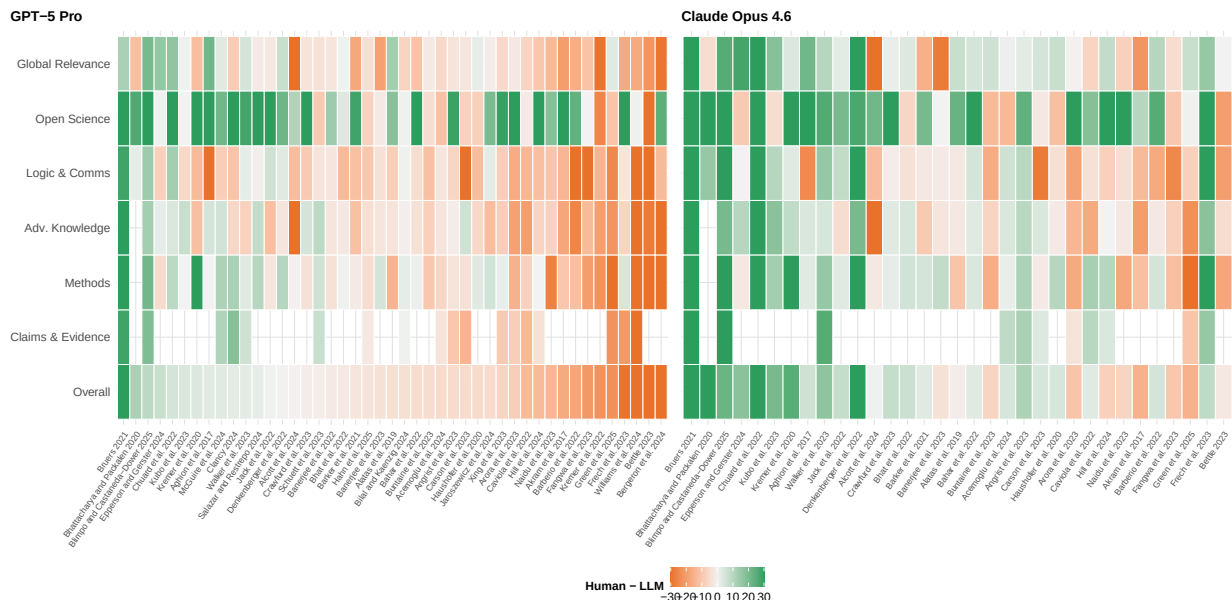


Papers where both curves slope the same direction indicate systematic human–LLM disagreement; papers where the two curves diverge indicate model-specific “taste” differences. The criteria-level Krippendorff’s alpha analysis in Table 3 (a chance-corrected reliability coefficient that generalises across varying numbers of raters, missing data, and any measurement level; see [Methods](#) for a formal definition) quantifies where human–LLM agreement approaches the ceiling set by human inter-rater variability.

Rating differences by paper and criterion. While the scatter plots and slopegraph summarise agreement on a single dimension, Figure 4 unpacks Human – LLM differences across all seven

criteria simultaneously. Each column is a paper, each row a criterion, and tile colour encodes the signed difference (human mean minus LLM midpoint). Green tiles indicate the human panel rated the paper higher; orange tiles indicate the LLM was more generous. Papers are sorted by GPT-5 Pro’s overall difference so that both panels share the same column order, making model-specific patterns immediately visible.

Figure 4: Human minus LLM rating difference for every paper (columns) and criterion (rows), comparing GPT-5 Pro (left panel) and Claude Opus 4.6 (right panel). Green tiles indicate the human mean was higher; orange tiles indicate the LLM rated the paper higher. Colour is clamped to ± 30 percentile points. Papers are sorted by GPT-5 Pro overall difference so the two panels are directly comparable.



Rows with uniformly green or orange tiles indicate papers where humans and the LLM disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases—for instance, if “Open Science” is green across nearly all papers, humans may systematically reward data-sharing practices more than the LLM does. Comparing the two panels reveals whether disagreement patterns are model-specific or shared across frontier LLMs; tiles that change colour between panels highlight papers where the models diverge from each other, not just from humans.

Qualitative critique comparison. Beyond numeric ratings, we compare the substantive critiques each model raises against the consensus issues identified by human experts. **?@fig-coverage-precision-main** plots coverage—the fraction of human-identified concerns that the LLM also raised in some form—against precision—the fraction of LLM-raised issues that correspond to a substantive human concern. These metrics are assessed by GPT-5.2 Pro acting as an independent judge (see [Methods](#) for the judging protocol).

Coverage and precision vary substantially across papers. Some papers achieve high scores on both dimensions, indicating strong alignment between human and LLM critiques, while others reveal the LLM missing key human concerns or raising issues absent from the expert consensus. Because these metrics are themselves LLM-assessed, they should be interpreted with the caveat that an LLM judge

may systematically over- or under-credit matches relative to a human annotator; manual validation through our annotation tool is underway and preliminary results are consistent with the automated scores.

Detailed paper-by-paper comparisons—including matched issue pairs with severity labels, structural difference tables, and per-evaluator breakdowns—appear in [Appendix B: Critiques & Key Issues](#). Extended quantitative analysis, including per-criterion correlations, bootstrap confidence intervals, mutual information, tier prediction accuracy, cost-quality trade-offs across all six models, and top disagreement cases, is reported in [Appendix A: Results Ratings](#). The full LLM reasoning traces and assessment summaries are available in [Appendix C: LLM Traces](#).

Discussion

Frontier LLMs achieve moderate-to-strong agreement with human expert evaluators, approaching the ceiling set by inter-rater variability among humans themselves. Correlations with human mean ratings are highest for reasoning-capable models (GPT-5 Pro, Claude Opus 4.6), though even lightweight models (GPT-4o-mini, Gemini 2.0 Flash) perform respectably at a fraction of the cost. Central compression—the tendency for LLMs to pull extreme ratings toward the middle of the scale—is the most consistent pattern across all models, likely reflecting alignment training that discourages confident extreme outputs. Qualitative coverage varies widely across papers: on some, the LLM captures nearly all consensus human concerns; on others, it misses key critiques or raises issues absent from the expert consensus.

Limitations. Several caveats temper these conclusions. Our sample comprises roughly 50 social-science papers specifically selected by The Unjournal for evaluation, not a random draw from the research literature; performance may differ in other fields or on less polished manuscripts. Human evaluations are themselves a noisy reference signal rather than ground truth, with substantial inter-rater variation that caps achievable agreement. We cannot fully rule out knowledge contamination: while we block extrinsic priors in the system prompt, the models’ training data may include fragments of these papers or related discussions. Alignment training likely contributes to score inflation and narrower credible intervals than humans provide. All LLM evaluations are single-run; aggregating across multiple runs or temperature settings could change the picture. Finally, the qualitative coverage and precision metrics are themselves LLM-assessed (GPT-5.2 Pro as judge), introducing a further layer of model dependence.

Implications. The cost structure is striking: lightweight models deliver evaluations at under a cent per paper, while reasoning-capable models cost several dollars—still orders of magnitude cheaper than human expert review. This suggests a practical tier structure in which cheap models could serve as rapid screening tools and expensive reasoners could provide deeper assessment where warranted. However, the qualitative gaps we observe—missed critiques, generic issues, and central compression of ratings—argue against full automation of peer review. AI evaluation appears most promising as a supplement: providing fast structured feedback, flagging potential concerns for human reviewers, and enabling systematic comparison across large paper sets that would be infeasible with human effort alone.

Future work. Several extensions follow naturally from this analysis. Content-swap bias tests following Pataranutaporn et al. (2025) would reveal whether LLMs show systematic biases based on author names or institutional affiliations. A journal-outcome prediction horse-race would compare human and LLM tier predictions against actual publication venues. Systematic prompt and model comparisons, including agent-based approaches, could identify configurations that reduce central

compression or improve qualitative coverage. Human enumerator validation—employing trained raters to independently assess whether LLMs identify the same consensus critiques—would ground the currently LLM-assessed coverage metrics. Out-of-time validation using papers entering The Unjournal’s pipeline would eliminate contamination concerns entirely. Finally, hybrid human-AI evaluation trials would test whether collaboration improves evaluation quality beyond either alone.

Methods

This chapter describes the data sources, evaluation pipeline, prompt design, and statistical methods used throughout. All Python code underlying the pipeline is preserved in the collapsible blocks below and can be executed to reproduce the evaluation runs.

Sample and human reference data. We draw on published evaluations from [The Unjournal](#), an open-access platform that commissions expert reviews of policy-relevant research without requiring journal submission. Each paper is typically assessed by two independent evaluators (occasionally one or three), who provide a written critique resembling a referee report together with quantitative ratings on seven criteria scored as percentiles (0–100) relative to “all serious research in the same area encountered in the last three years.”² Evaluators additionally predict the journal tier in which the work “should” and “will” publish, using a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable, 5 = top-5 journal). For each metric, evaluators report a midpoint (median of their belief distribution) and a 90% credible interval that expresses their epistemic uncertainty.

The sample comprises working papers spanning 2017–2025 in development economics, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. All papers have completed The Unjournal’s full evaluation process, meaning the authors received evaluations that have been publicly posted. For our analysis we extracted the individual evaluator scores, aggregated them by taking the arithmetic mean per paper per criterion, and retained the range of individual scores to characterise inter-rater spread.

LLM evaluation pipeline. We evaluate six frontier models spanning three providers: GPT-5 Pro and GPT-5.2 Pro (OpenAI, reasoning-capable), GPT-4o-mini (OpenAI, lightweight), Claude Sonnet 4 and Claude Opus 4.6 (Anthropic), and Gemini 2.0 Flash (Google). Each model receives the identical PDF file, system prompt, and output schema. We pass the PDF directly to the model’s native multimodal input rather than extracting text, preserving tables, figures, equations, and layout cues that ad-hoc scraping could mangle. A single API call per paper avoids hand-offs and summary loss from multi-stage pipelines.

The output is constrained by a strict JSON Schema enforcing the same nine fields that human evaluators complete: seven percentile metrics (each with `midpoint`, `lower_bound`, `upper_bound`) and two journal-tier predictions (each with `score`, `ci_lower`, `ci_upper`). Additionally, the model produces an `assessment_summary` of approximately 1,000 words that must precede scoring—a “think first, score second” protocol designed to ground numeric ratings in specific textual evidence. The extended schema used for our GPT-5.2 Pro focal run adds a `key_issues` array of concise, ranked issue statements for downstream critique comparison.

Agreement and reliability statistics. We quantify human–LLM agreement using several complementary measures. *Pearson’s r* captures linear association between paired ratings and is sensitive to proportional biases but not to constant offsets. *Spearman’s* measures rank-order agreement and is robust to outliers and non-linear monotone relationships. *Mean bias* (LLM minus human) indicates the direction and magnitude of any systematic offset. *Root mean squared error* (RMSE) and *mean absolute error* (MAE) measure typical prediction error in the original scale units; RMSE penalises large deviations more heavily. To contextualise human–LLM agreement we report *Krippendorff’s alpha* (α), a chance-corrected reliability coefficient that generalises across varying numbers of raters, accommodates missing data, and applies to any measurement level (nominal, ordinal, interval, or ratio). An α of 1 indicates perfect agreement, 0 indicates agreement

²The seven criteria are: *overall assessment*, *claims and evidence*, *methods*, *advancing knowledge and practice*, *logic and communication*, *open and collaborative science*, and *relevance to global priorities*. Full definitions are given in The Unjournal’s [guidelines for evaluators](#).

no better than chance, and values below 0 indicate systematic disagreement. We compute α_{HH} (among human evaluators only) as a ceiling: if human evaluators agree with each other at only $\alpha = 0.5$ on a given criterion, expecting an LLM to exceed that level would be unrealistic. We then report α_{HL} (between the human mean and each LLM) to assess how close machine ratings come to this ceiling. For the qualitative key-issue comparison, *coverage* denotes the fraction of human-identified issues that received a match score $\geq 30\%$ from the LLM judge, and *precision* denotes the fraction of LLM-generated issues that matched at least one human issue.

JSON Schema. The output schema enforces that every paper is scored on identical fields with identical types and bounds. Credible intervals are required (paralleling the human protocol) so that the model can express genuine uncertainty rather than suggest false precision.

System prompt design. The system prompt is assembled from modular components and concatenated before each API call. It opens with a role definition instructing the model to act as an expert research evaluator, followed by a debiasing block that explicitly prohibits use of author identity, institutional prestige, publication venue, or any extrinsic information—the model must base all judgments on the PDF content alone. A diagnostic-summary instruction requires the model to produce a roughly 1,000-word assessment identifying methodological, evidential, and interpretive issues *before* any scoring, implementing a “think first, score second” protocol intended to anchor numeric ratings in specific textual evidence.

Percentile ratings are anchored to the reference group “all serious research in the same area encountered in the last three years,” following The Unjournal’s [guidelines for evaluators](#). The prompt defines each of the seven criteria with emphasis on global priorities and practical relevance over pure academic novelty, mirroring the weight structure that human Unjournal evaluators are asked to apply.

Subsequent prompt components instruct the model on constructing 90% credible intervals—the smallest interval the evaluator believes is 90% likely to contain the true value—encouraging calibrated uncertainty rather than artificially narrow bounds. The prompt requests journal-tier predictions on a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable through 5 = top-5 journal), providing an externally verifiable benchmark for papers that eventually publish. A validation block then requires the model to verify internal consistency: numeric scores must align with the written assessment, credible intervals must be non-degenerate, and high or low ratings must be explicitly justified in the assessment summary.

The components above are concatenated into a single system prompt string before each API call:

Submission and collection. The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation. File IDs are cached by path, size, and modification time so that re-running on the same PDF reuses the previously uploaded file.

Each model receives the full PDF in a single reasoning call, avoiding hand-offs and summary loss from multi-stage ingestion. We submit one background job per paper to the OpenAI Responses API with “high” reasoning effort and server-side JSON-Schema enforcement, recording the response ID, model, file ID, status, and timestamps in a jobs index. No external sources or cross-paper material are retrieved; the evaluation is anchored entirely in the manuscript itself.

A polling loop then checks each job’s status and, for completed jobs, retrieves the raw JSON response object and writes it to disk alongside reasoning-trace metadata (token counts, reasoning summary).

Multi-model evaluation. To assess whether systematic biases or calibration differences emerge across architectures, we collect evaluations from six models spanning three providers. For OpenAI models that lack background-job or extended-reasoning support (e.g., GPT-4o-mini), we use a synchronous call variant:

Anthropic. Anthropic’s API accepts PDFs as base64-encoded document content rather than uploaded file IDs, and all calls are synchronous. We use Claude’s native PDF support to preserve the same multimodal evaluation approach:

Google. Google’s Gemini API accepts PDFs via a file-upload endpoint with MIME-type tagging:

A unified runner dispatches evaluations across all configured providers and writes per-model output directories:

Focal run with key-issue extraction. For a subset of 14 papers with rich human critiques, we ran an extended evaluation using GPT-5.2 Pro. The schema for this focal run adds a `key_issues` array—a ranked list of concise issue statements identifying the most important methodological, interpretive, or evidential concerns—alongside the standard metrics. This structured output enables direct comparison between machine-generated and human-identified issues.

Key-issue comparison with human critiques. To validate how well the LLM identifies substantive concerns, we compare its `key_issues` output against human expert critiques drawn from The Unjournal’s Coda database. These human critiques—produced by paid domain experts and synthesized by evaluation managers—provide a high-quality but noisy reference standard for issue identification; accordingly, we treat them as a comparative signal rather than ground truth.

The comparison proceeds in two stages. First, we parse and align the data sources: LLM key issues are extracted from the focal-run JSON responses, while human critiques are drawn from a manually curated markdown document pairing each paper’s machine and expert assessments. Second, we use an LLM judge (GPT-5.2 Pro with schema-enforced structured output) to systematically assess the degree of alignment between each human-identified issue and the set of machine-generated issues, producing issue-by-issue match scores, coverage estimates (fraction of human issues captured), and precision estimates (fraction of LLM issues that are genuinely substantive).

The comparison pipeline takes a manually curated markdown file pairing each paper’s LLM issues with human critiques and produces two structured outputs: a parsed JSON dataset (`key_issues_comparison.json`) and an LLM-assessed alignment report (`key_issues_comparison_results.json`) containing per-paper coverage, precision, matched-pair explanations, and an overall quality rating. Together, these outputs feed the qualitative analysis presented in [?@sec-results-critiques](#).

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Results: Ratings

We compare quantitative ratings from 6 frontier LLMs (GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6, Gemini 2.0 Flash) against human expert reviews from [The Unjournal](#). 45 papers have both human and LLM evaluations, enabling direct comparison across all seven percentile criteria and two journal-tier predictions.

We submit each PDF through a single, schema-enforced API call. The system prompt mirrors The Unjournal rubric, blocks extrinsic priors (authors, venue, web context), and requires strict JSON: a ~1k-word diagnostic summary plus numeric midpoints and 90% credible intervals for every metric.³ Each numeric field carries a midpoint (model’s 50% belief) and a 90% credible interval; wide intervals indicate acknowledged uncertainty.

Cost and token usage. Token costs are computed from API-reported input/output tokens (GPT-5 Pro runs with `reasoning = high`).

Table 1: Token usage and estimated cost per model

Model	Papers	Avg input	Avg output	Avg reasoning	Total cost	Cost/paper
Claude Opus 4.6	45	116,930	1,509	—	\$84.02	\$1.867
GPT-5 Pro	60	38,516	6,400	4,717	\$57.71	\$0.962
GPT-5.2 Pro	14	45,143	2,889	707	\$11.91	\$0.850
Claude Sonnet 4	16	113,798	727	—	\$5.64	\$0.352
GPT-4o-mini	60	71,581	470	—	\$0.66	\$0.011
Gemini 2.0 Flash	60	18,084	1,024	—	\$0.10	\$0.002

Total cost across all 255 LLM evaluations: **\$160.03**. Reasoning-capable models (GPT-5 Pro, GPT-5.2 Pro, Claude Opus 4.6) cost substantially more per paper than their lighter counterparts (GPT-4o-mini, Gemini 2.0 Flash), raising practical questions about cost-quality trade-offs for deployment at scale.

Agreement metrics. We assess agreement using Pearson r (linear correlation), Spearman (rank-based, robust to outliers), mean bias (LLM – Human; positive = LLM rates higher), and RMSE/MAE (error magnitude). Full definitions of all statistics appear in [Methods](#).

Table 2: Agreement between each LLM and human mean ratings (overall criterion)

Model	N	Pearson r	Spearman	Mean bias	RMSE	MAE
Claude Opus 4.6	33	0.567	0.477	-7.6	17.2	12.4
GPT-5 Pro	45	0.342	0.517	+6.4	15.4	10.3
Gemini 2.0 Flash	45	0.342	0.313	-5.3	14.5	11.7
GPT-5.2 Pro	9	0.333	0.471	-2.7	19.0	14.7
GPT-4o-mini	45	0.135	0.116	+0.0	20.2	13.1

³0–100 percentiles relative to a reference group of ~‘all serious work you have read in this area in the last three years’, overall (global quality/impact), claims_evidence (claim clarity/support), methods (design/identification/robustness), advancing_knowledge (contribution to field/practice), logic_communication (internal consistency and exposition), open_science (reproducibility, data/code availability), and global_relevance (decision and priority usefulness). Journal tiers: (0–5, continuous) capture where the work should publish vs. will publish.

Table 2: Agreement between each LLM and human mean ratings (overall criterion)

Model	N	Pearson r	Spearman	Mean bias	RMSE	MAE
Claude Sonnet 4	11	0.020	0.055	+0.2	9.5	6.9

Table 2 reports correlation and error metrics between each LLM and the human mean overall rating. Positive mean bias indicates the model rates papers higher than humans on average; negative bias indicates harsher ratings. These correlations are bounded above by human inter-rater noise—since the human “mean” is itself noisy, no model can achieve a perfect correlation.

Human baseline context. To interpret human-LLM agreement, we compare against human-human agreement as an upper bound. If human evaluators only agree with each other at $\alpha = 0.5$, expecting LLMs to exceed that would be unrealistic. Table 3 shows α_{HH} (Krippendorff’s alpha among human evaluators) alongside α_{HL} (between the human mean and each LLM).

Table 3: Human-human vs Human-LLM agreement by criterion (Krippendorff’s α)

Criterion	α_{HH}	α_{HL} (GPT-5 Pro)	α_{HL} (GPT-5.2 Pro)	α_{HL} (GPT-4o-mini)	α_{HL} (Claude Sonnet 4)	α_{HL} (Claude Opus 4.6)	α_{HL} (Gemini 2.0 Flash)
Overall	0.55	0.25	0.36	0.14	0.06	0.47	0.25
Claims	0.42	0.20	0.26	0.19	0.21	0.26	0.18
Methods	0.54	0.29	0.07	0.03	-0.22	0.39	0.16
Adv.	0.20	0.18	0.20	0.13	0.06	0.32	0.07
Knowl- edge							
Logic & Comms	0.33	-0.04	-0.19	-0.00	0.26	0.09	-0.03
Open Science	0.03	-0.06	-0.61	-0.16	0.25	-0.18	0.03
Global Rele- vance	0.33	0.25	0.51	0.04	0.06	0.18	0.17

Where α_{HL} approaches α_{HH} , the LLM is performing at or near the ceiling set by human inter-rater variability. Criteria where α_{HH} is already low (indicating genuine evaluator disagreement) naturally cap how well any model can do. Extended agreement analysis—including CI coverage, bootstrap confidence intervals, mutual information, and per-criterion correlations—is available in the supplementary material.

Criteria-level patterns. Figure 5 displays the mean rating each evaluator type assigns across the seven criteria. Criteria where all models and humans converge (similar colour intensity) suggest robust agreement on that dimension; criteria with divergent columns indicate systematic differences in how humans and LLMs weigh that aspect of quality.

Additional models. Figure 6 shows overall ratings for the two lightweight models—GPT-4o-mini and Gemini 2.0 Flash—against human mean ratings. These models cost 50–200× less per paper than reasoning-capable models (see Table 1) but may sacrifice rating accuracy.

Figure 5: Mean rating by evaluator type (rows) and criterion (columns), averaged across all matched papers. Tile colour scales from light (lower mean) to vivid orange (higher mean), clamped to 40–80. Criteria where all rows show similar colour intensity indicate broad agreement; divergent columns flag criteria-level systematic differences.

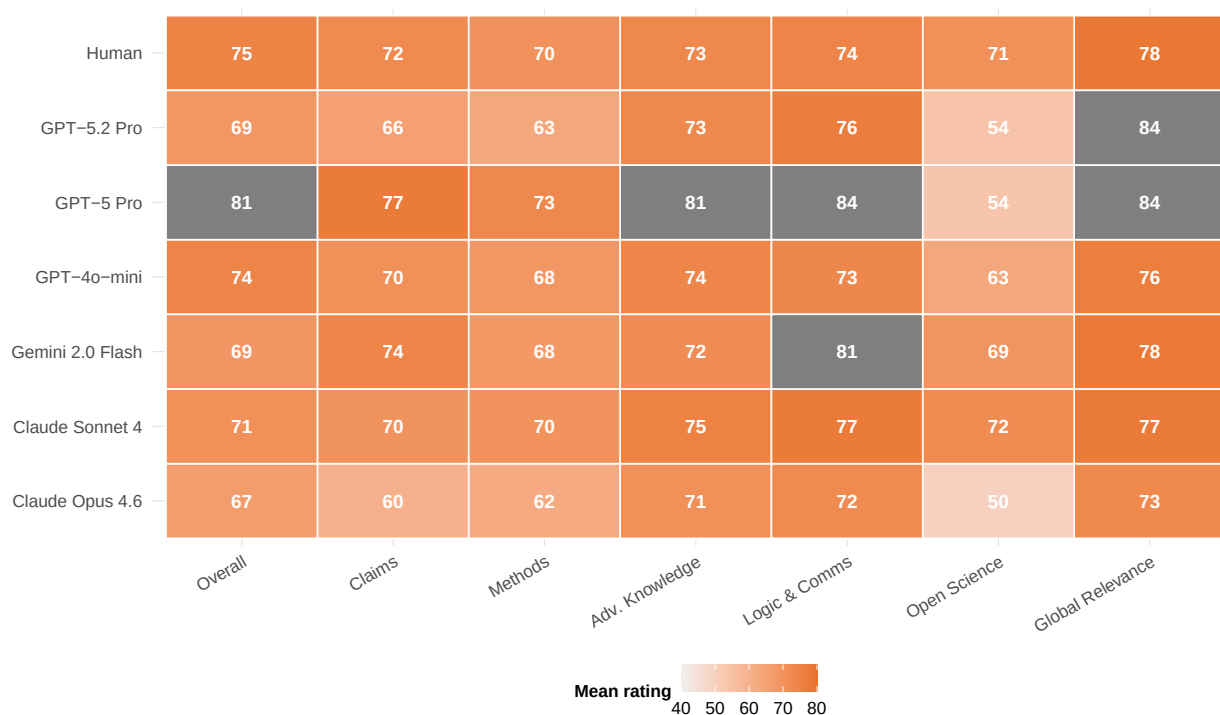
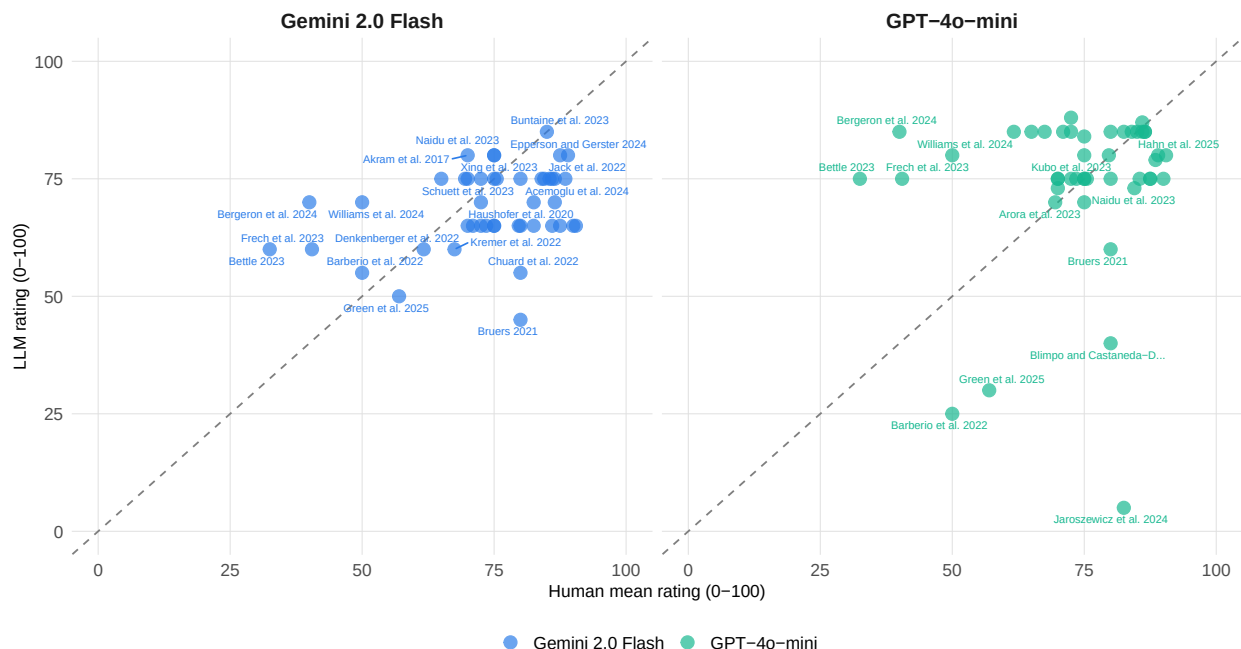


Figure 6: Human mean overall rating (x-axis) vs LLM overall rating (y-axis) for the two lightweight models GPT-4o-mini and Gemini 2.0 Flash. Each point is one paper; the dashed diagonal is the identity line. These models cost 50–200 $\times$ less per paper than reasoning-capable models (see cost table above).



For per-paper scatter plots, rank comparisons, and the gap heatmap showing Human – LLM differences by paper and criterion, see the [main Results chapter](#). Extended reasoning traces are available in [Appendix C: LLM Traces](#).

Results: Critiques & Key Issues

This chapter reports preliminary results from comparing LLM-identified key issues against human expert critiques. The analysis uses LLM-based assessment (GPT-5.2 Pro as judge) to measure coverage and precision; human validation of these alignment scores is underway.

Warning

This section is under active development. Coverage and precision metrics below are LLM-assessed; manual annotation is in progress.

Data sources. We compare GPT-5.2 Pro key issues (structured output from the focal evaluation run, January 2026) against human expert critiques curated from The Unjournal’s internal tracking database, where evaluation managers synthesize evaluator feedback using severity labels (“Necessary”, “Optional but important”, “Unsure”).

We matched **14 papers** with both GPT-5.2 Pro key issues and human expert critiques.

LLM-based assessment. The comparison was assessed using GPT-5.2 Pro, which evaluated coverage (what proportion of human concerns GPT identified) and precision (whether GPT issues are substantive).

- **Coverage:** Proportion of *consensus* human issues that have any LLM match (match quality $\geq 30\%$)
- **Weighted Coverage:** Mean match quality across all human issues (treating unmatched issues as 0%)
- **Precision:** Proportion of LLM issues that match a human concern

i Caveat on Precision

“Precision” only reflects whether LLM issues match the *consensus* human issues curated in Coda—those prioritized by evaluation managers as the most important concerns. Many LLM-identified issues may have been noted by one or more individual human evaluators but were not included in this curated set. A low precision score does not necessarily mean the LLM raised irrelevant concerns; it may simply reflect issues that weren’t prioritized in the consensus summary.

i Interpretation Guide

- **Coverage:** Percentage of human-identified issues that GPT also captured (in some form). Higher = GPT missed fewer human concerns.
- **Precision:** Percentage of GPT issues that are substantive rather than generic or spurious. Higher = GPT’s critiques are more targeted.
- **Overall Rating:** Qualitative assessment (Excellent/Good/Moderate/Poor) based on both coverage and precision metrics.

The coverage-precision figure above shows substantial variation across papers: some papers achieve high coverage and precision, while others show the LLM missing key human concerns or raising issues not in the expert consensus. Paper-by-paper comparisons with matched issue pairs, structural difference tables, and the manual annotation tool are available in the [full online version](#).

Supplementary Material

Related Work

AI practical research applications

Luo et al. (2025) survey LLM roles from idea generation to peer review, including experiment planning and automated scientific writing. They highlight opportunities (productivity, coverage of long documents) alongside governance needs (provenance, detection of LLM-generated content, standardizing tooling) and call for reliable evaluation frameworks.

Korinek (2025) considers the specific potential for using AI in economic research, offering “working examples and step-by-step code” to help economists use agents for literature reviews, econometric code, fetching and analyzing economic data, and coordinate complex research workflows. He advocates economists using AI tools to boost productivity and “to understand [these tools] capabilities and limitations.” He calls for human researcher oversight “to ensure [these tools] pursue the right objectives and maintain research integrity”. He outlines a range of current limitations and open questions (outlined [here](#)), including hallucinations, mistakes compounding through multi-agent workflows and “brittleness to small variations in prompts”. At the higher level, he argues “the questions that will matter most—what problems deserve investigation, how should we evaluate trade-offs, what constitutes progress—will remain irreducibly human.”

Mitchener, Yiu, et al. (2025) present Kosmos, “an AI scientist for autonomous discovery”. However, evaluations reveal important limitations: Kosmos was found to be only 57% accurate in statements that required interpretation of results, and “identifying valuable discoveries Kosmos made is a time-intensive process that relies on human scientists with significant domain expertise.”

AI peer review, error detection, and research evaluation

Son et al. (2025) evaluates LLMs on 83 papers paired with 91 confirmed significant scientific errors. Their strongest (OpenAI o3) model identified only 21.1% of these errors, with only 6.1% precision—domain experts found most false positives were hallucinations or student-level misconceptions.

Eger et al. (2025) provide a broad review of LLMs in science and a focused discussion of AI-assisted peer review. They argue: (i) peer-review data is scarce and concentrated in CS/OpenReview venues; (ii) targeted assistance that preserves human autonomy is preferable to end-to-end reviewing; and (iii) ethics and governance (bias, provenance, detection of AI-generated text) are first-class constraints.

Dycke, Kuznetsov, and Gurevych (2023) present NLPEER, a “multidomain corpus” of more than 5000 papers and 11k review reports from 2012 to 2022 from five different venues.

Zhang and Abernethy (2025) propose deploying LLMs as quality checkers to surface critical problems instead of generating full narrative reviews. Using papers from WITHDRARXIV and an automatic evaluation framework that leverages “LLM-as-judge,” they find the best performance from top reasoning models but still recommend human oversight.

Pataranutaporn et al. (2025) asked four nearly state-of-the-art LLM models to consider 1220 unique papers drawn from 110 economics journals excluded from the training data of current LLMs. They find general alignment between LLM ratings and RePEC journal rankings, mixed results on detecting AI-generated papers, and evidence that LLMs show biases towards named high-prestige male authors and elite institutions.

Zhang et al. (2025) use AI conference paper data and LLM agents for pairwise manuscript comparisons, finding this approach outperforms traditional rating-based methods for identifying high-impact papers by citation metrics, though with some evidence of biases favoring established research institutions.

AI research taste and potential to guide research

A key motivation for this work stems from broader claims about AI’s potential to accelerate scientific discovery. If AI systems are to contribute meaningfully to research—not just as tools but as evaluators and potential collaborators—we need to understand their “taste”: what they value, what they miss, and whether their priorities align with human expert judgment and real-world research impact. This connects to broader debates about the returns to science in the presence of technological risk (Clancy 2023).

Extended Rating Analysis

The main text reports six core exhibits: cost overview, human-vs-LLM scatter plot, agreement metrics, human baseline (Krippendorff’s alpha), criteria-level heatmap, and gap heatmap. The following extended analyses are preserved in the project’s git history and available in earlier rendered versions:

- **Krippendorff’s alpha by criterion:** Inter-rater agreement treating human mean and each LLM as separate raters
- **CI coverage analysis:** Share of human mean ratings falling within LLM 90% credible intervals
- **Bootstrap 95% CIs:** Nonparametric confidence intervals for Pearson correlations
- **Mutual information:** Information-theoretic measure of human-LLM rating dependence
- **Per-criterion correlations:** Pearson and Spearman by evaluation criterion and model
- **Tier correlation dumbbell plot:** How criteria predict journal tier judgments for humans vs. LLMs
- **Top disagreement cases:** Papers with largest human-LLM rating differences
- **Prompt comparison:** Legacy vs. updated GPT-5 Pro prompt effect on ratings

Extended Critique Analysis

The main text reports coverage-precision scatter and summary statistics. The following extended analyses are available in the [full online version](#):

- **Paper-by-paper issue comparisons:** Interactive collapsible panels showing matched issue pairs, unmatched human/LLM issues, and detailed discussion for each paper
- **Aggregate statistics:** Total issue counts by severity, human vs. LLM issue distributions
- **Observable structural differences:** Format, structure, attribution, and length comparisons
- **Manual annotation tool:** Browser-based UI for systematic concordance labeling

The per-paper LLM evaluation traces below are only available in the [online HTML version](#).